

Chapter

Millet: Traditional “Poor Man’s” Crop or Future Smart Nutri-Cereals?

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Abstract

Millet represents a diverse group of cereal crops of significance to sub-Saharan Africa and globally. However, they remain a set of crops with limited attention and priority paid to them with paucity of information on their genetic diversity and sustainable use. Existing knowledge on millets with respect to cultivation, health, and nutritional benefits, and contribution to sustainable environmental management, and use is mainly attributed to traditional indigenous knowledge held by rural folks in different regions of the continent. The emergence of other cereal staples, however, led to millets losing their place as an important crop limiting their use to a “famine” crop with production occurring on smallholdings among the marginalized poor. This threatens interest, patronage, conservation and use to sustainably and fully exploit the potential of millets for the benefit of society. Intertwined with tradition and culture, millets in sub-Saharan Africa and elsewhere nonetheless hold great promise to contribute to food security, revitalize and diversify diets, improve farmer livelihoods, resilience, and adaptation to climate change. This chapter discusses the importance of millets, challenges to production, contribution to nutrition and health, traditional knowledge and products, novel and non-traditional products, contribution to resilience and climate change, and diversity of available genetic resources.

Keywords: millets, small grains, cereals, nutrition, health, climate resilience, value-addition, utilization, finger millet, pearl millet, sub-Saharan Africa

1. Introduction

Millets are an important set of cereal crops belonging to the family Poaceae. They have a wide distribution and are associated with cultures and food festivals among indigenous peoples of Africa and Asia. Of significance is the association of millets with food and nutrition security in marginalized arid and semi-arid regions of Sub-Saharan Africa (SSA) and elsewhere. A diversity of millets are cultivated in SSA, Asia and the Americas including but not limited to pearl millet [*Pennisetum glaucum* (L.) R. Br.], foxtail millet (*Setaria italica* (L.) P. Beauvois), little millet (*Panicum sumatrense* Roth.ex. Roem. & Schult.), Japanese barnyard millet [*Echinochloa esculneta* (A. Braun) H. Scholz], kodo

millet [*Paspalum scrobiculatum* L.], proso millet [*Panicum miliaceum* L.], finger millet [*Eleusine coracana* (L.) Gaertn], Indian Barnyard millet [*Echinochloa frumentacea* Link], fonio or ach (*Digitaria exilis* (Kippist) Stapf), black fonio (*Digitaria iburua* Stapf), guinea millet (*Brachiaria deflexa*), and tef [*Eragrostis tef* (Zucc.) Trotter] [1]. Within the SSA context, four millets, that is, pearl millet, finger millet, fonio and teff are prevalent and of significance depending on geographical location – that is, east, central, southern and west Africa. In these regions of sub-Saharan Africa, millets contribute to the dietary energy and protein needs for at least 130 million people with significant impact on their food and nutrition security [1]. Of the four millets cultivated in Africa, pearl millet and finger millet stand out and have a wider distribution and global importance [2].

Pearl millet ranks sixth globally and is cultivated in areas with limited rainfall mainly in arid and semi-arid areas of SSA. Finger millet is majorly grown in more humid areas of SSA with its center of origin assumed to be in East Africa and more specifically, Uganda [3]. Traditionally, millets are cultivated by vulnerable groups including women and youths and are used as food security crops during times of scarcity. Furthermore, millets receive limited attention and investment for research and development, innovation and value addition, policy and political support. Oftentimes, referred to as a “poor man’s crop”, the importance and contribution of millets to the population in SSA and globally remains unnoticed. This is despite the innate, unexploited, biochemical, structural and nutritional characteristics potential of millets in food, feed, energy and industrial sectors. To fully harness and exploit millets for improved socio-economic and environmental benefit, increased effort towards development to address specialty market needs is imperative to drive demand and productivity. To achieve this within the context of climate change and variable weather patterns, self-reliance in food production in SSA will rely on low input agriculture in poor production environments.

2. Millets production trends in SSA

Women and youth comprise the primary producers of millets in SSA and together with other cereals provide the primary source of dietary energy for more than 962 million people. Nonetheless, production remains low, approximately 2.5 times below that of the developing world average characterized by subsistence level agriculture on small land holdings measuring 0.3–5.0 ha. A myriad of factors may contribute to this observation including extreme weather events (floods and drought), poor agronomic practices, lack of improved adapted genetic resources resistant to biotic and abiotic stresses, access to quality seed, traditional low yielding varieties, absence of appropriate smart and precision agriculture technologies for development and production of new adapted material, lack of well defined value chains for cultivated cereal crops, labour, bird damage and loss of essential crop biodiversity.

Production areas for millets span arid and semi-arid regions in east, south and west Africa. This has significant impact on rural socio-economic development, food security, feed security, energy security, health, and environmental sustainability for resource constrained persons in SSA. The major producing countries in sub-Saharan Africa include Nigeria, Niger, Burkina Faso, Senegal, Chad, Sudan, Mali, Uganda and Zimbabwe. In all of these countries, millet production is characterized by low input agricultural practices [4] in marginal lands with depleted soil fertility, high temperatures and drought [5]. Production volumes varied from 3038 to 3,886,079 tonnes per year from 2016 to 2020 with Niger producing the highest quantity of millet, followed by Nigeria, and Mali (**Figure 1**). Comparatively, production trends for millets in east

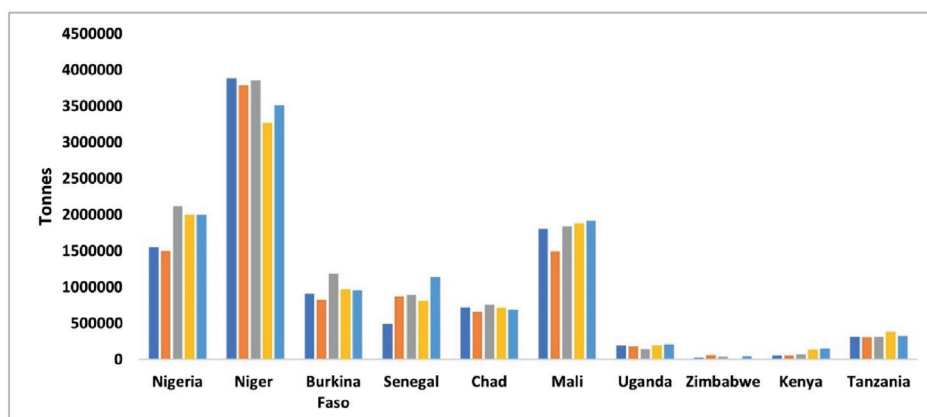


Figure 1.
 Production quantity of millet from millet-producing countries in sub-Saharan Africa for the years 2016–2020 (FAOSTAT 2021).

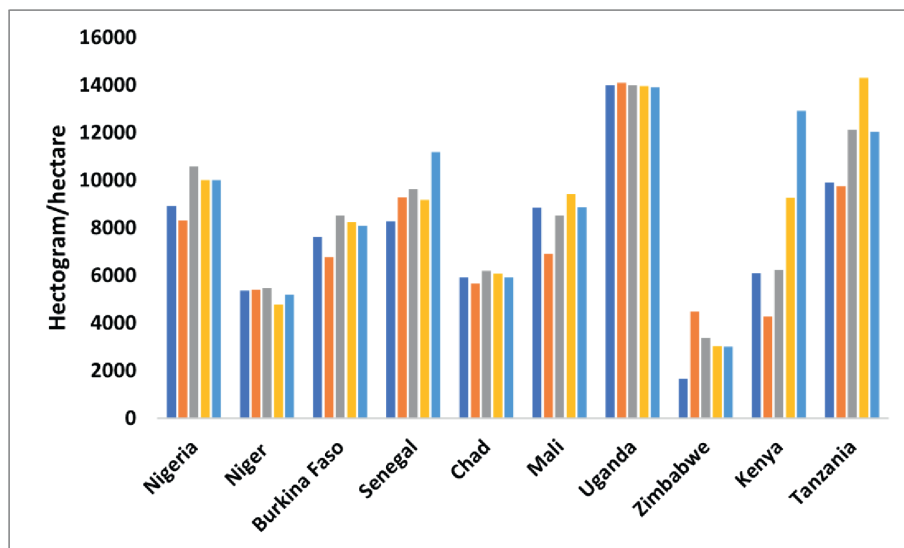


Figure 2.
 Millet productivity in different millet-producing countries in sub-Saharan Africa for year, 2016–2020 (FAOSTAT 2021).

and southern Africa is low, with Tanzania ranked as the highest producer in the region followed by Uganda. Considering yield per unit area, East Africa has the highest millet productivity, with the highest yield (13,919–14,094 kg/ha) reported in Uganda from 2016 to 2020 (**Figure 2**). Other countries with high millet productivity in SSA include Tanzania, Senegal and Nigeria.

3. Why millets?

Despite the pivotal role and contribution to household food security, millets have over the years lost their place in today’s cereal production systems and diets. This

threatens interest, patronage, conservation and use to sustainably meet food security, farmer livelihoods and climate resilience for rural and urban communities alike. A number of factors could have contributed to the loss in popularity of millets relegating them to cultural food festivals and crops associated with famine hence, food for the poor. Furthermore, the decreasing trend in popularity and consumption of millets may be attributed to the availability of introduced cereals including maize, wheat, and rice, their associated public distribution system, easy methods of processing and preparation for consumption. The requirement for traditional knowledge and special skill mainly available among rural folks in preparing millet dishes and non-availability of ready-made value added products in the market further deter wider use of millets as food. Furthermore, the concept of the green revolution, promotion and adoption of alternate cereals such as hybrid maize varieties, access to credit, government support on maize prices and marketing subsidies also contributed to the decline in production of millets in the region [6]. Other challenges include the menace of quelea birds, drudgery and rising labour costs in production, emergence of fast foods and changing consumer preferences, widespread biotic stresses including striga infestation [7], diseases such as blast and downy mildew [7] and lack of remunerative market prices. Intertwined with tradition and culture, millets in sub-Saharan Africa and elsewhere nonetheless hold great promise to contribute to food security, revitalize and diversify diets, improve farmer livelihoods, adaptation to marginal soil conditions and climate resilience. The re-emergence in popularity of millets in SSA and elsewhere is due to the comparative advantages the crop offers in light of its potential contribution to climate change, urbanization and growing interest in nutrition and healthy foods. Millets adapt well to a wide range of climatic and marginal environments hence their compatibility with climate-smart agriculture interventions. Furthermore, the crops have a long post-harvest storage life and are seldom affected by storage pests [2].

3.1 Millets contribution to nutrition and health

Millets are a nutritionally valuable group of crops. Besides energy, millets provide a good source of quality protein and micronutrients. Pearl millet ranks highest in protein content (14.5%) among the millets, comparable to wheat (14.4%). Furthermore, pearl millet is associated with a more complete amino acid profile with valuable amino acids including methionine and tyrosine which are lacking in some millets [8, 9]. Both pearl millet and finger millet are rich in branched-chain amino acids (Leucine, isoleucine and valine) which have essential roles in protein synthesis, metabolic and regulatory processes [10, 11]. Finger millet possesses essential vitamins such as Vitamin A and B complex, water and fat-soluble vitamins [12]. Essential micronutrients present in pearl millet and finger millet include zinc, iron, calcium, sodium, potassium and magnesium [11]. Dietary consumption of millets facilitates vasodilation attributed to magnesium and potassium, which contributes to keeping the human body blood pressure at optimum [13, 14]. Consumption of finger millet has been reported to reduce plasma triglycerides, which helps reduce the risk of cardiovascular disease [15]. Furthermore, millets are rich in polyphenols and tannins, which contribute to reducing the risk for different kinds of cancer among populations consuming millets in their diets [16]. This observation is attributed to high radical scavenging or quenching activity in finger millet [17, 18]. Millets are a rich source of antioxidants such as curcumin, quercetin and ellagic acid which act as immune boosters, and important in the health and wellbeing of children and young adults [19].

As such, millets are of significance to integrated feeding and nutrition programs aimed at managing malnutrition and hidden hunger in SSA [20].

Millets have considerable health benefits given that they are reported to be gluten free [21]. As such, they are considered suitable for coeliacs. Coeliac disease is a syndrome characterized by damage to the mucosa of the small intestine caused by ingestion of gliadins and glutenins associated with wheat, rye and barley [21]. The prevalence of coeliac disease is estimated as high as 1:266 [21]. In the United States of America, persons with coeliac disease are estimated to be approximately 3 million. On the contrary, there are limited or no cases of coeliac disease in SSA. This may be because coeliac disease goes un-noticed and or cases are not reported or, the reliance of the rural population the majority of whom rely on millets. Furthermore, traditional knowledge across SSA has it that millets are natural nutraceuticals with meals often prepared and offered to persons with diabetes. Despite the association of millets with nutraceutical attributes even among local folks, there still remains a paucity of medical research of millets on human health with respect to the effects of phytochemicals and other components of the small grain cereals.

3.2 Traditional knowledge and products from millets

Across sub-Saharan Africa, traditional uses of millets are limited in scope mainly for food or local beverages and or brew. Part of the limitation in use is also attributed to the requirement of special skill in preparing and processing millets, which knowledge is not common place. This tends to limit the demand for millets, particularly in urban areas where consumers have access to alternate processed value-added foods. Promotion and food enhancement of millets can contribute to improve the nutritional status and livelihoods of poor marginalized communities. Traditional technologies such as fermentation, malting, soaking, milling/grinding, cooking, roasting and popping are common place in SSA [22]. Each of these processing technologies improve the nutritional value, functional qualities and sensory properties of millets. For example, soaking reduces anti-nutritional compounds such as trypsin inhibitors, phytate, tannins, phytic acids and flavonoids, thus increasing the bioavailability of minerals such as zinc and iron [23]. Germination and malting facilitate hydrolysis of proteins and starch, making amino acids and sugars readily available in processed products, improves digestibility and availability of vitamins and minerals [24, 25].

Preparation and processing of traditional millet products varies by region and cultures within SSA. The product range may include but not be limited to creamy millet drink (*Kanu-zaki* in Nigeria), fermented millet porridge or gruel (*Ogi* in Nigeria, *Ben-saalga*, *Ben-kida* in Burkina Faso, *Malwa* or *Ajon* in Uganda), unfermented millet porridge or gruel (*Bushera* in Uganda, *Uji* in Kenya or Tanzania, *Togwa* in Tanzania), and millet bread or dough (*Kalo* in Uganda, *Tehobal* and *gappal* in Burkina Faso) (**Table 1**). In Ethiopia, gluten-free leavened breads (*injera*) may be prepared from millets [21].

With improvements and access to technology, across regions in SSA, various other value-added millet products have been formulated using several components such as yoghurt, milk, ginger, chili, other cereal blends, baobab pulp or leaves, tamarind juice, fruits such as pineapple and spices. The combination improves both the taste/sensory attributes as well as the nutritive value of the formulated millet products. It is worth noting that some nutrients such as vitamins may be reduced or lost during the processing of millets into other ready-to-drink products [26]. The diversity of millet products especially in West Africa shows that the utilization

Region	Country	Product	Description	Recipe/composition
West Africa	Nigeria	<i>Masa</i>	Fried cake	Millet, yeast, yoghurt, sugar, salt, onion
		<i>Kunu-zaki</i>	Creamy millet drink	Millet, sugar, germinated rice, cloves, ginger, potato, milk
		<i>Fura</i>	Boiled millet balls	Millet, cloves, ginger, chili pepper
		<i>Ogi</i>	Fermented millet porridge	Millet, water, fermentation culture
	Burkina Faso	<i>Massa</i>	Millet pancake	Millet, rice, Maize Tô, oil, water
		<i>Ben-saalga</i>	Fermented millet gruel	Millet, ginger, black pepper, mint, water
		<i>Ben-kida</i>	Fermented millet gruel	Millet flour, millet granules, ginger, pepper, mint, water
		<i>Millet gruel</i>	Millet gruel for Northern region	Millet, sugar, curd, cooking oil, salt, baobab pulp, water
		<i>Zom-kom</i>	Crude millet beverage	Millet, sugar, ginger, mint, Tamarind leaves, tamarind fruits, pineapple powder, canafra, water
		<i>Millet pase</i>	Thick millet dough	Millet flour, water, tamarind juice
		<i>Tounka</i>	Bran millet beverage	Millet flour, millet bran, sugar, salt, solid potash, water
		<i>Tehobal</i>	Millet dough	Millet flour, curd cheese, sugar, water
		<i>Gappal</i>	Millet dough with milk	Millet flour, curdled milk or yoghurt, sugar, salt, water
		<i>Millet couscous</i>	Granular steamed food	Millet flour, water, salt, baobab leaf or okra powder
East Africa	Senegal	<i>Thiakry</i>	Millet couscous topped with mangoes, raisin, coconuts & nuts	Millet, salt, water, milk, yoghurt, vanilla, cinnamon, chopped pistachios
	Mali	<i>Toh (Oro Dja)</i>	Traditional food of the Dogon people	Husked millet, water, dried okra, dried fish, dried baobab leaves, fermented hibiscus seeds
	South Sudan	<i>Medeeda</i>	Millet porridge with yoghurt	Millet, water, peanut, sugar, lime juice and yoghurt
	Cameroon	<i>African Millet parties</i>	Millet cookies	Millet, raisin, water, onion, celery seeds, soy source
	Uganda	<i>Bushera</i>	Millet porridge	Millet flour, water, milk, sugar
		<i>Kalo</i>	Millet bread	Millet flour, cassava flour, water
		<i>Mahwa/ajon</i>	Millet beer	Germinated millet, water, sorghum
		<i>Uji</i>	Millet porridge	Millet flour, water, milk, sugar
East Africa	Kenya/ Tanzania	<i>Togwa</i>	Millet gruel	Germinated finger millet, maize flour, water

Region	Country	Product	Description	Recipe/composition
Southern Africa	Zimbabwe	<i>Mahewu</i>	Millet malt drink	Millet malt, water
		<i>Sadza rezviyo/ Isitshwala samabele/ Inyawuthi</i>	African millet pap	Millet, water
		<i>Masvusvu</i>	Millet mash	Malted millet, water

Table 1.
 Traditional millet-based products and their composition from different sub-Saharan countries.

and demand for millet can be improved in SSA, which will directly benefit farmers, processors and other value chain actors. Across different African countries, similar millet products are given different names. For example, millet beer is named differently in different areas such as *Malwa* or *Ajon* (Uganda), *Merissa* (Sudan), *Chikokivana* or *Ithothotho* (Zimbabwe), *Chagga* (Tanzania), *Tchouk* (Togo) and *Dolo-Djioula* (Burkina Faso). There may be variations in the preparation of millet beer in these regions, however, standardization of this process offers an opportunity for promoting millet beer in a wider area, increasing the production and utilization of millets in SSA.

3.3 Novel and non-traditional millet products

Over the years, advances in food processing technologies and cereal sciences in general have presented opportunity for the development of novel and non-traditional products from diverse cereals including maize, wheat, rye, rice and barley. These advancements can be of significant value in the development of millets in the quest to improve and support their re-adoption, production and commercialisation in the face of growing health concerns and climate change.

Tortillas, snack foods, noodles, starch breads, flour breads and additives, cakes, cookies, and pasta are examples of novel products that have been successfully produced from other cereals. One hundred percent millet bread and/or confectionery products remains the main challenge. However, this challenge could be ameliorated with additives and enzymes to improve product quality.

As opposed to traditional African opaque beer, malting and brewing with millets to produce lager and stout, often referred to as clear beer, has not been conducted on a large, commercial scale. This is because brewing processes with millets have not been extensively investigated and are only at an experimental and/or piloting stage.

However, significant traditional beverages from millets exists diverse in recipe by culture and region within SSA. More recently, the Coca cola company launched the Ades global brand of plant-based beverages. In Uganda, the Coca cola company added Ades Nutri-Bushera to its product portfolio, which offers an opportunity to boost millet production, productivity and income for populations living in marginal arid and semi-arid areas of SSA [27]. Through cultivar selection and crop improvement, millets can be used to produce non-food products including bioethanol and other bio-industrial products such as bioplastics, thus contributing to rural circular bioeconomy in marginal areas [21].

3.4 Contribution of millets to climate change adaptation and resilience

Climate change remains a significant challenge to agriculture in SSA presenting with unpredictable weather, limited and erratic rainfall. However, millets hold promise to secure food and nutrition security for the approximately 1.17 billion inhabitants in SSA. Governments across SSA have put in place agriculture sector, Development Strategy and Investment Plans that seek to increase household incomes and improve food security by enhancing agricultural production and productivity, achieve economic growth through industrialization, reduce greenhouse gas emissions and become middle-income countries by the year 2050. In line with the African Union (AU) Agenda 2063, millets have the potential to contribute to the aspirations – “A prosperous Africa, based on inclusive growth and sustainable development” and “An Africa whose development is people driven, relying on the potential offered by people, especially its women and youth and caring for children”.

The importance of millets as a climate-smart cereal in contributing to climate change adaptation, mitigation and resilience cannot be underestimated for a number of reasons. Farmers in SSA are familiar with its cultivation and is easy to propagate. Millets are adaptable to diverse environments and tolerates heat and drought stress. With decreasing water supplies, increasing population growth and their requirement for limited inputs during growth, millets represent an important set of genetic resource for future human use. The annual growth cycle of millets (2.5–4 months) does not tie up land that can be used for other uses and, can be rotated or intercropped with other food crops ensuring food security, improved soil integrity and minimizes disease and pest epidemics. The root system of millets facilitates below-ground carbon sequestration providing rapid carbon sink and lowering greenhouse gas emissions.

4. Millets genetic resources and improvement

Millets biodiversity is an invaluable resource to African agricultural production systems to help address unpredictable climatic conditions, household nutrition, and socio-economic challenges in SSA. This can be achieved through crop diversification, conservation, and sustainable use integrated with indigenous knowledge. About 133,849 cultivated germplasm of small millets are conserved in gene banks worldwide the majority being conserved in Asia (64.4%), followed by Africa (13.8%), and Europe (13.5%) [28]. More than 22,211 pearl millet and 35,382 finger millet have been collected and consolidated in different gene banks with the largest collection at ICRISAT. Of these, a few have been utilized as breeding/research material [29]. Despite these numbers, conservation of millets in genebanks in SSA and globally is generally compared with other crop species. It is plausible this might be due to the low research and priority accorded to millets and associated land races leading to loss in biodiversity. In Uganda for instance, pearl millet biodiversity has been reduced or even lost with farmers citing damage from birds as the primary reason for abandoning the crop despite its importance. To prevent further loss of existing, available diversity, it is imperative that concerted efforts in supporting collection, conservation and use of available millets’ diversity is carried out before we lose them forever.

The year 2023 has been declared the “International Year of Millets” by UN General Assembly in recognition of its importance as a climate smart food crop with to contribute to alleviation of malnutrition, hunger and poverty. The latest UN projection suggests that the world population could grow to around 9.7 billion in 2050, with

Sub-Saharan Africa contributing over 50% of this population growth [30]. The inter-governmental panel on climate change warned that the global food supply is already jeopardized [31]. Therefore, resilience of millets to climate and weather variations makes it a suitable future smart Nutri-cereal. To achieve this, efforts in crop improvement and processing technologies are needed to generate “smart super millets” with specialty traits targeting industrial applications to increase demand, productivity and subsequently popularity of millets. A number of methods and tools are available including conventional crop improvement and marker assisted breeding approaches. Target traits of interest for crop improvement will include diseases and pest, drought and heat stress, nutrition quality for macro- and micronutrients, yield and water use efficiency. In livestock communities, biomass would be an important trait for dual purpose millet varieties to provide food and feed security.

More recently, the speed breeding approach has been used as a strategy to reduce the time frame for variety development in crops such as wheat. Speed breeding can be integrated with tools including high throughput phenotyping, genotyping by sequencing (GBS), genome editing technologies such as CRISPR-Cas9, and biochemical analyses to develop climate smart millets. Progress by the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in millets improvement, has focused on pearl millet and finger millet [28] with development of millet hybrids high in yield, with high absorbable nutrients. A clear example is the work done by HarvestPlus, a CGIAR challenge program leading global efforts in the development of micronutrient-rich crop varieties. The program started in 2003, focusing on the biofortification of millet, bean, cassava, rice, sweet potato, wheat and maize [32, 33], contributing to the alleviation of malnutrition due to iron and zinc deficiency affecting 60% and 30% of the world population respectively [34]. Evidence of the impact of biofortified pearl millet with absorbable iron concentration of 60 ug/g has been documented [23]. So far, a total of 11 high iron pearl millet varieties, 9 single cross and 2 open-pollinated varieties have been released [33], only one variety has been released in Africa (in Niger). Efforts should be made to evaluate all 11 varieties in different millet producing countries for a decentralized selection of the best millet varieties for nutrition programs to address different health problems caused by iron deficiency [35, 36].

Open source web-based platforms are available with high-computing capabilities and comprehensive breeding management and analysis software such as Usegalaxy (<http://usegalaxy.org>), Breedbase (<http://breedbase.org>), integrated breeding platform (<http://www.integratedbreeding.net>), and Excellence in breeding platform (<http://excellenceinbreeding.org>) to facilitate accelerated crop improvement in low and middle-income countries. These efforts are complemented by genotyping services at regional hubs such as SEQART Africa that offers sequencing services in support of national breeding programmes. The availability of such service facilitates molecular breeding such as diversity studies, genome-wide association studies for unraveling of the genetic architecture of key traits, genomic selection and gene discovery for fast tracked breeding and increased genetic gain in different crops.

5. Conclusions and future perspectives

The re-emergence of millets as an important set of crops not just for SSA but globally demonstrates their potential to serve as alternate and/or supplement cereals along side major staples – maize, rice, wheat, and barley. Nonetheless, significant efforts still remain to bring millets at par with other cereals given they remain virtually

unresearched and their potential unexploited. From a crop improvement perspective, understanding genetic diversity within and between millets defines the first step in their development for effective and sustainable utilization. To guide the process of millets improvement, farmers with traditional indigenous knowledge, industry and other stakeholders in the value chain need to participate as partners providing a platform for co-creation and development of value-added products subsequently driving consumer demand, incomes and rural economic development. To enhance the rate of genetic gain and improvement in millets, adoption of tools including the use of artificial intelligence, high throughput phenomics and genomics are imperative to assure timeliness and efficiency in selection. It is worthwhile noting that, climate-ready millets in themselves are not sufficient to achieve increased productivity, food security, rural incomes, resilience and adaptation to climate change. Rather, identifying specialty attributes in improved millets genetic resources that meet niche market needs coupled with circular economy approaches will address limitations of value chains and create supplier linkages for millions of small-scale farmers and consumers across SSA. To complement this, as with other crops, increase in productivity of millets can become a reality if stakeholders in SSA can adopt precision agriculture technologies including but not limited to the use of unmanned aerial vehicles to monitor crop health, growth and physiology; integrating laser scare crows alongside other technologies for bird management, farm management software, farm mechanization and postharvest processing.

To attract the support of different authorities including government to promote research, development and commercialisation of millets across Africa, a look into millets beyond just contributing to resilience and household food security is imperative. Deliberate efforts need to demonstrate the contribution of millets to health and nutrition, rural employment, household incomes, rural and national economies, and sustainable environment management. Such efforts may include but not be limited to policies support on partnerships on co-development of high value millet products with large customer bases; integrating gender equality aspects in the promotion of millets taking into consideration the contribution of rural women in genetic resource management, household food security, and their role in subsistence agriculture.

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Conflict of interest

The authors declare no conflict of interest.

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